Graph Convolutional Networks for Protein Function Prediction

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Why Predict Protein Functions?

R1
Biological Gap

Genomic sequencing produces protein sequences faster than experimental function determination

R2
Cost Reduction

Computational predictions are cheap and fast compared to lab experiments

R3
Drug Discovery

Knowing what a protein does helps us design drugs to target it when it malfunctions

R4
Systems Biology

Individual proteins work together in complex networks. Our GCN approach captures these interactions

Task Formulation

Multi-Label Node Classification on Graphs

Input:

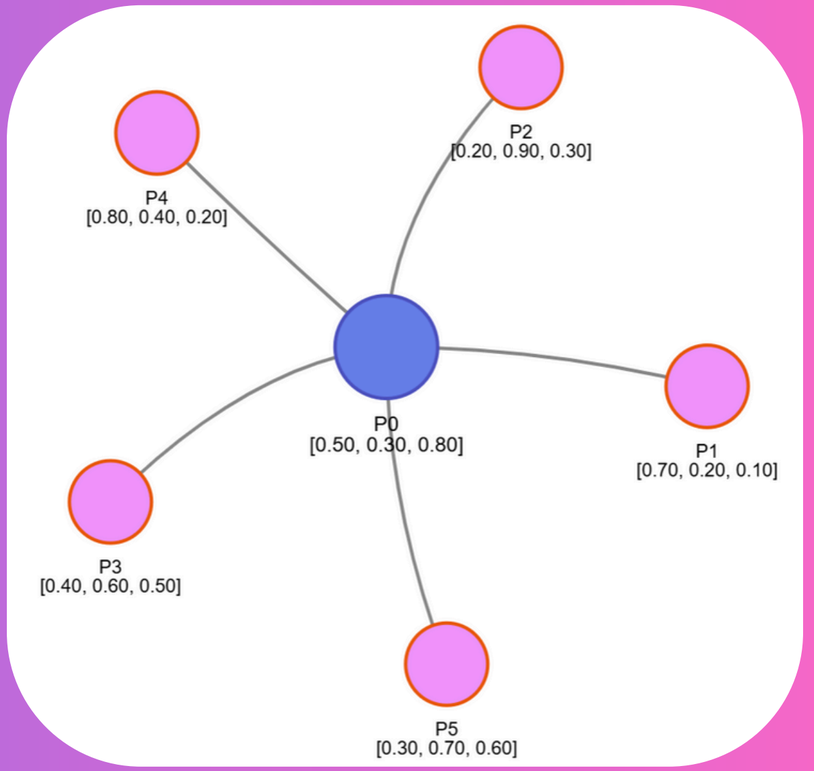
Node features: 50 biological descriptors per protein

Graph structure: Protein-protein interaction network

Output:

Multi-label predictions: 121 Gene Ontology terms per protein

Each protein can have multiple functions simultaneously



Dataset Characteristics

Class Distribution:

Some functions are common (appear in many proteins)

Many functions are rare (few proteins have them)

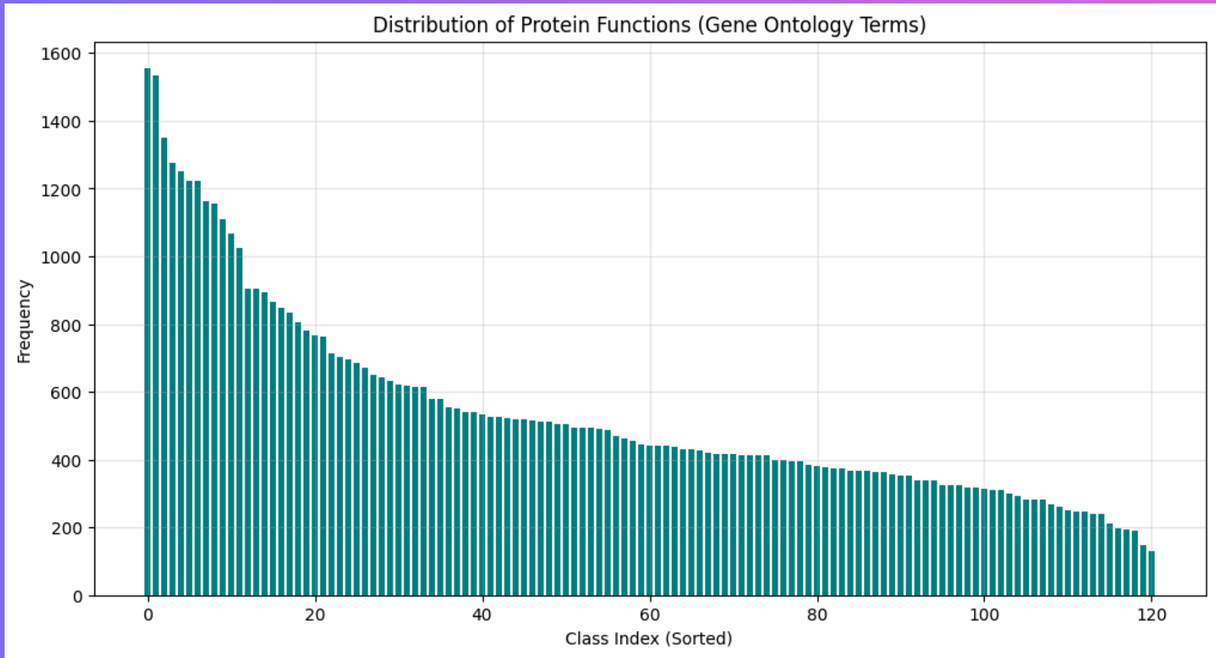
Typical long-tail distribution in biological datasets

Implications:

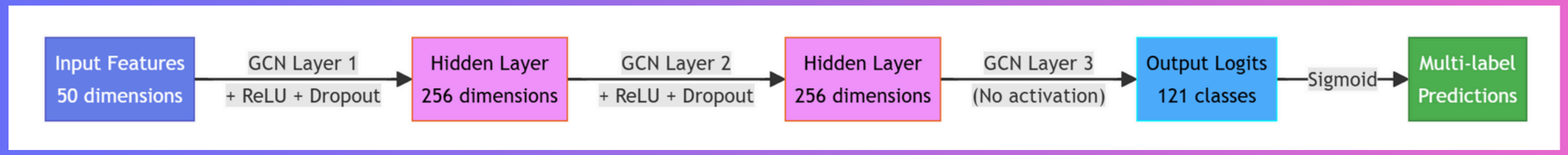
Accuracy alone is misleading

Need specialized metrics that handle imbalance

Model must learn both frequent and rare functions



Solution Architecture





2-4 GCN layers

Loss Function

Binary Cross Entropy with Logits Loss

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{121} \left[y_{ij} \cdot \log(\sigma(\hat{y}_{ij})) + (1 - y_{ij}) \cdot \log(1 - \sigma(\hat{y}_{ij})) \right]$$

Implementation Details:

Sigmoid activation applied within loss function

No final layer activation in network architecture

Can incorporate class weights for imbalance handling

Evaluation metric

Micro-Averaged F1 Score

Justification:

Aggregates contributions of all classes

Gives equal weight to each instance (protein-function pair)

Better for imbalanced data than macro-averaging

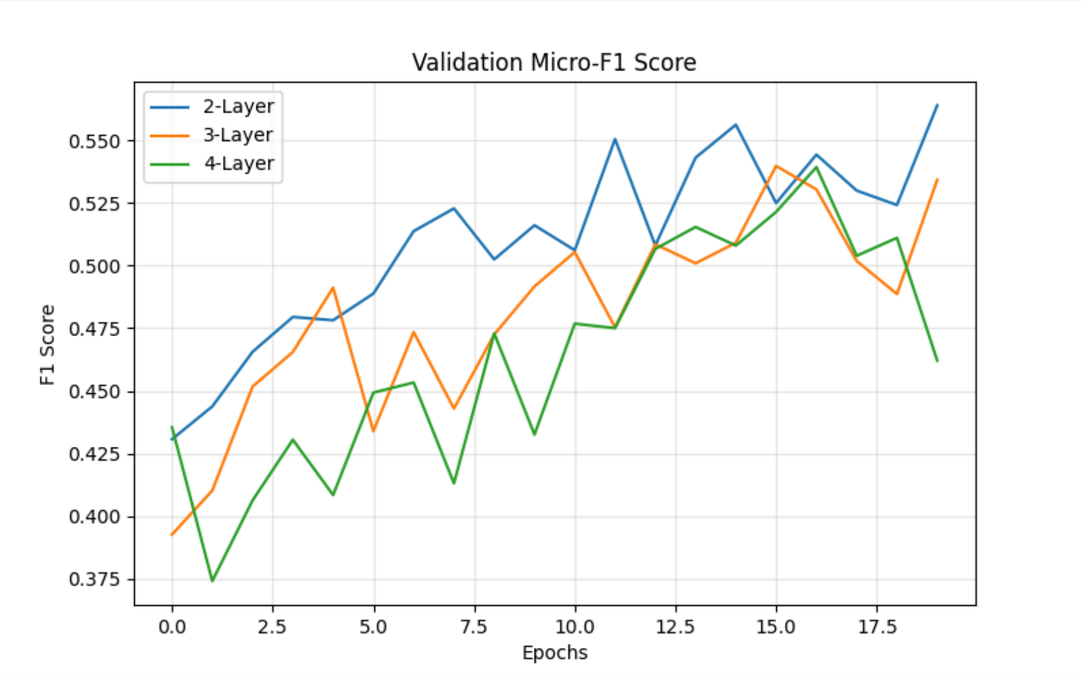
$$\text{Micro-F1} = \frac{2 \times \text{Micro-Precision} \times \text{Micro-Recall}}{\text{Micro-Precision} + \text{Micro-Recall}}$$

$$\text{Micro-Precision} = \frac{\sum \text{TP}}{\sum \text{TP} + \sum \text{FP}}$$

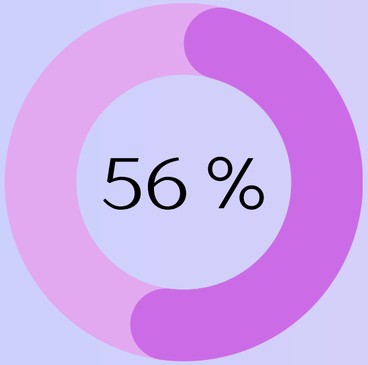
$$\text{Micro-Recall} = \frac{\sum \text{TP}}{\sum \text{TP} + \sum \text{FN}}$$

Experiments with number of layers

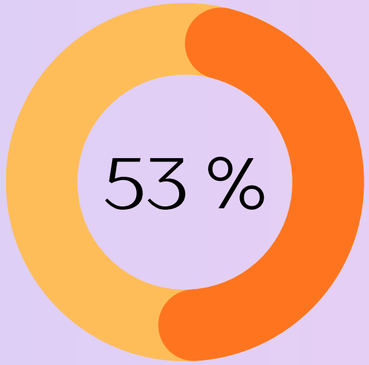




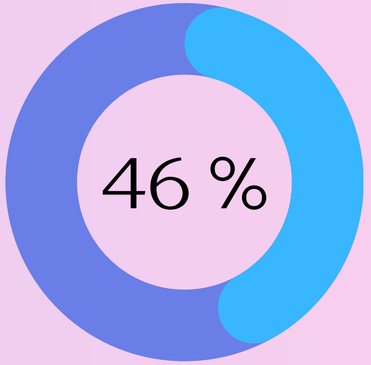
Final results



2-Layer



3-Layer



4-Layer

Why Does a 2-Layer Network Perform Best?

01

Over-smoothing

02

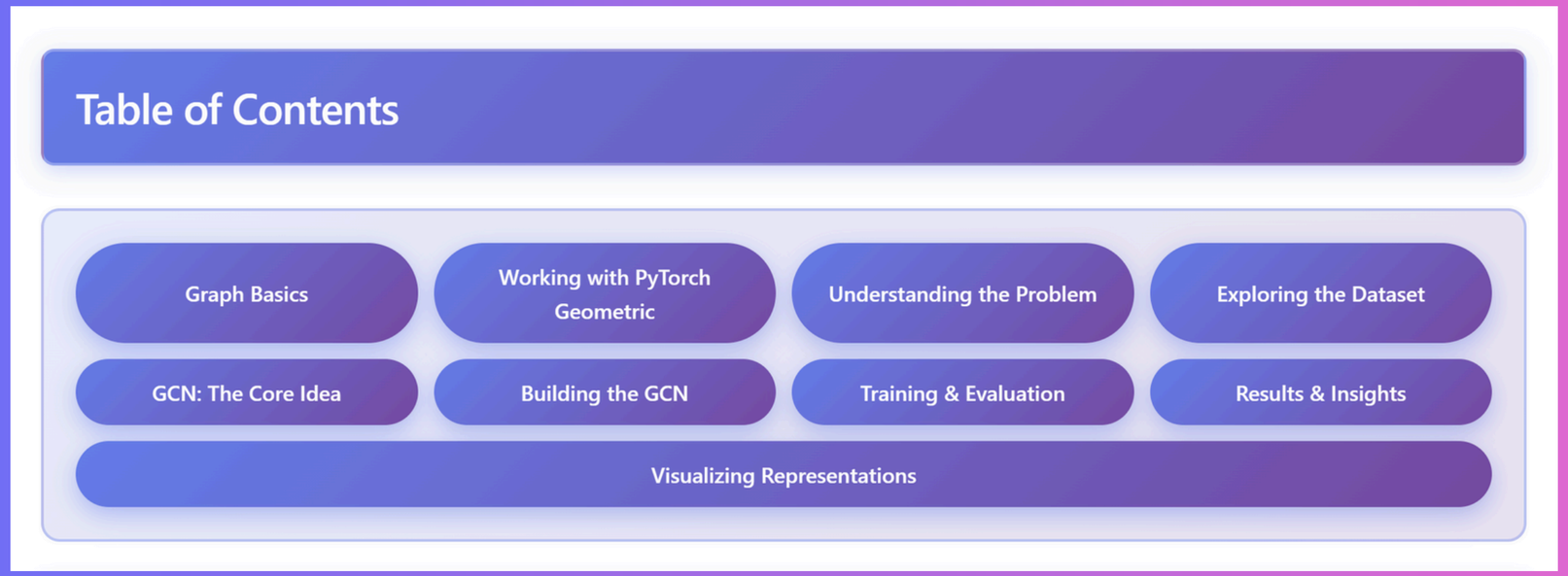
Task-specific optimal depth

03

Small receptive field

Product Demo

Blog post: https://aisylufattakhova.github.io/



Colab notebook: link

Thank you for attention